

Geometric Foundations of Data Analysis I

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September 23, 2026

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1 Introduction

Data analysis and more broadly Data Science is a vast and important field within Computer Science and Mathematics. At the core, the goal is to make sense of data, which can be measurements, survey results, behavior patterns, etc. Often this data comes to us in a very “high dimension”. That is, there are so many variables that it is impossible to visualize, and even in low dimensions, it may not be clear what the best conclusion is based on the analysis.

A few references seem to agree that the *total data* on all computers is something like **10^{23} bytes** or about **100 zettabytes**. While all of this data is not concentrated in one organization, we still require highly sophisticated tools to make sense of a huge amount of data. My goal with this course is that you will have a solid foundation with some standard tools. From this bedrock one could explore more sophisticated methods of data analysis more easily.

We will consider four key topics:

1. Least Squares Fitting,
2. Principal Component Analysis,

3. k -Means and Hierarchical Clustering,
4. Nearest Neighbors and the Johnson–Lindenstrauss Theorem

We will be working with the assumption that **the data we care about is preserved by orthogonal and linear transformations**. This is not true with all data—for example, one should not take (proper) linear combinations of people. However, for data like grams of different kinds of food, this is completely plausible. This assumption will not always be necessary, but we will just keep this in mind.

Half of our time will be spent bringing these ideas to life and getting our hands dirty. We will be working with Jupyter Notebooks to build familiarity with the concepts we will discuss. This will be done using Python and standard data analysis packages like Pandas.

2 Least squares fitting

This method of analysis is both simple and powerful. Like with most things in mathematics, without some good guiding examples, we can get lost in the formulas.

2.1 Build up

Suppose we have the following data as seen in Figure 2.1. (Maybe a company produces parts once per month in lots that vary in size depending on demand. We write i for the production run, x_i for the lot size, and y_i for the hours of labor.)

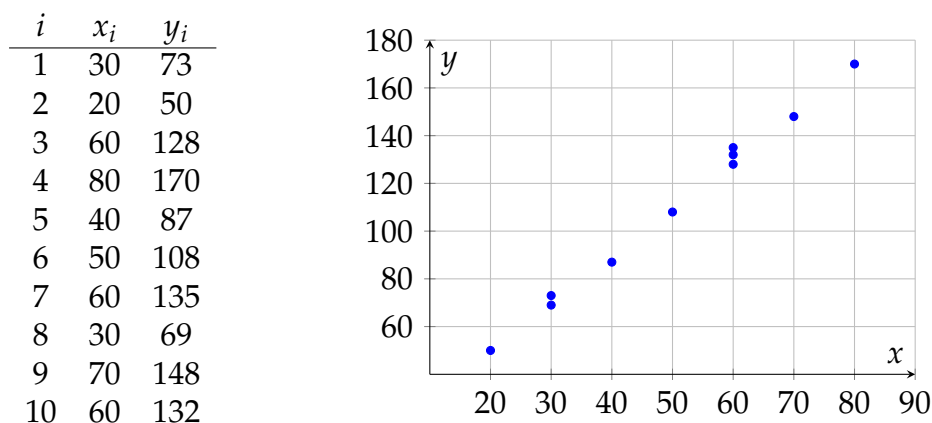


Figure 2.1: Data points exhibiting a linear relationship.

It seems clear from the plot that the data fits a geometric pattern—there is a linear phenomenon. If we try to find the line that is somehow closest to all the data points, we might draw something like in Figure 2.2.

Although the line is not a perfect fit, it seems to tell us something about the relationship between the lots size and the amount of hours.

Questions:

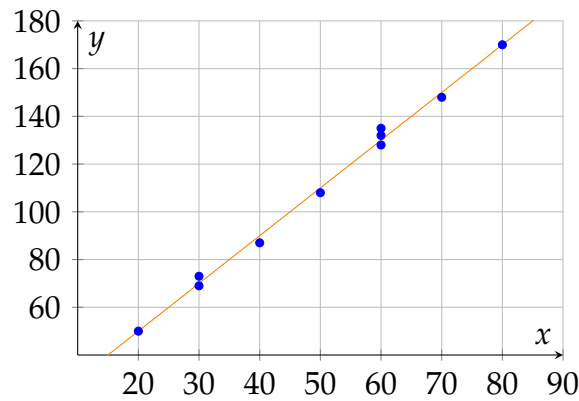


Figure 2.2: Line of best fit with data points.

- What makes this line “better” than alternatives?
- How are we quantifying “better”?
- Why are we using a line?

We will answer the first question later, probably. Let us consider the question of quantifying “better”. Least squares fitting is all about minimizing the squares of differences between the line and the actual data points. We will make this precise very soon.

2.2 Line of best fit

We know that the equation of a non-vertical line has the form

$$y = b_0 + b_1x.$$

If we had n data points of the form (x_i, y_i) , we could choose b_0 and b_1 to minimize the following sum

$$S(b_0, b_1) = \sum_{i=1}^n (y_i - (b_0 + b_1x_i))^2.$$

We can even solve for these values. Since S is a function in terms of b_0 and b_1 , all possible minima occur when the partial derivatives of S are 0. In other words, the minima arise as values (b_0, b_1) such that

$$\begin{aligned} \frac{\partial S}{\partial b_0} &= -2 \sum_{i=1}^n (y_i - (b_0 + b_1x_i)) = 0, \\ \frac{\partial S}{\partial b_1} &= -2 \sum_{i=1}^n (x_i y_i - x_i(b_0 + b_1x_i)) = 0. \end{aligned}$$

These equations are linear equations, so we can solve for these with techniques from linear algebra. This means, we need to solve two equations in the unknown

b_0 and b_1 :

$$\begin{aligned} nb_0 + b_1 \sum x_i &= \sum y_i, \\ b_0 \sum x_i + b_1 \sum x_i^2 &= \sum x_i y_i. \end{aligned} \quad (2.1)$$

Using the data from our example in Section 2.1, we have

$$\sum x_i = 500, \quad \sum y_i = 1100, \quad \sum x_i^2 = 28400, \quad \sum x_i y_i = 61800.$$

Thus, the equations we need to solve are

$$\begin{aligned} 10b_0 + 500b_1 &= 1100, \\ 500b_0 + 28400b_1 &= 61800, \end{aligned}$$

which yield $b_0 = 10$ and $b_1 = 2$. Going back to the context of the initial problem: this solution tells us that by increasing the lot size by one, we expect to increase the labor hours by two.

2.2.1 Written as matrices

Let us write the equations in (2.1) with matrices. This might seem like overkill at this stage, but it will set us up nicely to generalize. Let

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad B = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}. \quad (2.2)$$

Therefore, the equations in (2.1) are equivalent to the single matrix equation:

$$X^t X B = X^t Y. \quad (2.3)$$

If $X^t X$ is invertible, then $B = (X^t X)^{-1} X^t Y$.

2.3 In class exercises pt. I

1. (a) With X , Y , and B as defined in (2.2), show that

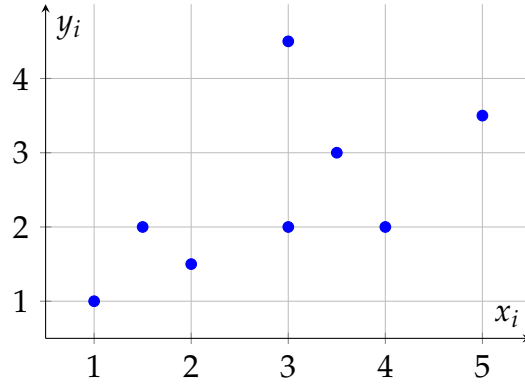
$$X^t X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad X^t Y = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}.$$

- (b) Show that $X^t X B = X^t Y$ is equivalent to (2.1):

$$\begin{aligned} nb_0 + b_1 \sum x_i &= \sum y_i, \\ b_0 \sum x_i + b_1 \sum x_i^2 &= \sum x_i y_i. \end{aligned}$$

2. Find a least squares fitting line to the following data and draw in the line:

i	x_i	y_i
1	1.0	1.0
2	2.0	1.5
3	3.0	2.0
4	1.5	2.0
5	3.5	3.0
6	3.0	4.5
7	4.0	2.0
8	5.0	3.5



(Round b_0 and b_1 to the nearest half integer.)

2.4 Hyperplane of best fit

Now we go to the general case. Suppose we have $p - 1$ independent variables and 1 dependent variable, where $p \geq 2$. We assume we have n data points of the form

$$(x_{i1}, x_{i2}, \dots, x_{i,p-1}, y_i) \in \mathbb{R}^p.$$

The least squares fitting for these data is a hyperplane of the form

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_{p-1}x_{p-1}.$$

To solve for the values b_i , we do as we did before. We define matrices

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,p-1} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,p-1} \end{pmatrix}, \quad Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad B = \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_{p-1} \end{pmatrix}.$$

As before, the values we want are given by the equation

$$X^t X B = X^t Y. \quad (2.4)$$

2.5 Why Equation (2.4) works

The heart of least squares is (Euclidean) distance. The distance between two points $x = (x_1, \dots, x_p)$ and $y = (y_1, \dots, y_p)$ in $\mathbb{R}^p$ is

$$d(x, y) = \|x - y\| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_p - y_p)^2}.$$

For a vector $v = (v_1, \dots, v_p) \in \mathbb{R}^p$, the **length** of v is

$$\|v\| = d(0, v) = \sqrt{v_1^2 + v_2^2 + \dots + v_p^2}.$$

Recall that the dot product of two (column) vectors u and v is

$$u \cdot v = u^t v = u_1 v_1 + u_2 v_2 + \cdots + u_p v_p.$$

Thus, the length of v is $\|v\| = \sqrt{v \cdot v}$; in other words $\|v\|^2 = v \cdot v$. In addition, if $u \cdot v = 0$, we say that u and v are **orthogonal** (or perpendicular).

Definition 2.1. Let U be a subspace of a vector space V . Define $U^\perp$, called the **orthogonal complement** (or **perp-space**) of U , to be

$$U^\perp = \{v \in V \mid \forall u \in U, u \cdot v = 0\}.$$

Lemma 2.2. Let U be a subspace of $V = \mathbb{R}^p$.

1. $U^\perp$ is a subspace of V .
2. $(U^\perp)^\perp = U$.
3. $U \cap U^\perp = \{0\}$.
4. $\dim(U) + \dim(U^\perp) = p$.

Proof. Try to prove this yourself. If you need help, you can read [5, Sec. 6.2]. $\square$

The goal of least squares is to *minimize distance*; more specifically to minimize $\|Y - XB\|$. Note that the column vector Y has entries that are the *actual* y_i values, and the column vector

$$XB = \begin{pmatrix} B \cdot (1, x_{11}, x_{12}, \dots, x_{1,p-1}) \\ B \cdot (1, x_{21}, x_{22}, \dots, x_{2,p-1}) \\ \vdots \\ B \cdot (1, x_{n1}, x_{n2}, \dots, x_{n,p-1}) \end{pmatrix} = \begin{pmatrix} b_0 + b_1 x_{11} + b_2 x_{12} + \cdots + b_{p-1} x_{1,p-1} \\ b_0 + b_1 x_{21} + b_2 x_{22} + \cdots + b_{p-1} x_{2,p-1} \\ \vdots \\ b_0 + b_1 x_{n1} + b_2 x_{n2} + \cdots + b_{p-1} x_{n,p-1} \end{pmatrix}$$

Therefore, $\|Y - XB\|$ is the square root of a sum of squares of the form

$$y_i - b_0 + b_1 x_{i1} + b_2 x_{i2} + \cdots + b_{p-1} x_{i,p-1}.$$

Hence minimizing $\|Y - XB\|$ is the same as minimizing $\|Y - XB\|^2$, which is a sum of *squares*. To achieve this, we first need to discuss orthogonal decompositions.

Proposition 2.3. Let U be a subspace of $V = \mathbb{R}^p$, and let $v \in V$. We can write v uniquely as

$$v = v_U + v_{U^\perp},$$

where v_U is in U and $v_{U^\perp}$ is in $U^\perp$. Additionally, v_U is the unique vector of U closest to v .

Proof. Let $m = \dim(U) \leq p$. By Lemma 2.2, $\dim(U^\perp) = p - m$. From linear algebra, we can extend bases. Thus, if $\{e_1, e_2, \dots, e_m\}$ is a basis for U and $\{e_{m+1}, e_{m+2}, \dots, e_p\}$ is a basis for $U^\perp$, then $\{e_1, e_2, \dots, e_p\}$ is a basis for V . Therefore, there exist unique $c_1, c_2, \dots, c_p \in \mathbb{R}$ such that

$$v = c_1 e_1 + c_2 e_2 + \dots + c_p e_p.$$

Let

$$v_U = c_1 e_1 + c_2 e_2 + \dots + c_m e_m, \quad v_{U^\perp} = c_{m+1} e_{m+1} + c_{m+2} e_{m+2} + \dots + c_p e_p,$$

which by construction are contained in U and $U^\perp$, respectively. This proves the existence and uniqueness.

Suppose $w \in U$. We will show that

$$\|v - v_U\| \leq \|v - w\|,$$

and hence v_U is the closest vector to v . We write

$$v - w = (v - v_U) + (v_U - w).$$

Since U is a subspace and $v_U, w \in U$, their difference $v_U - w$ is in U . Since $v_{U^\perp} = v - v_U \in U^\perp$, we have that $v - v_U$ is orthogonal to $v_U - w$. Now apply the Pythagorean Theorem to the (right!) triangle with sides $v_U - w$ and $v - v_U$ and hypotenuse $v - w$:

$$\|v - w\|^2 = \|v - v_U\|^2 + \|v_U - w\|^2.$$

Since lengths are non-negative, $\|v - w\|^2 \geq \|v - v_U\|^2$, as needed. $\square$

Proposition 2.4. *The minimal distance $\|Y - XB\|$ is achieved when*

$$X^t X B = X^t Y.$$

Proof. Consider the subspace $U = \{Xu \mid u \in \mathbb{R}^p\}$ of $\mathbb{R}^n$, and observe that our desired solution XB is contained in U . Since $\|Y - XB\|$ is minimal, we must have that the vector $Y - XB$ is orthogonal to all vectors contained in U by Proposition 2.3. In other words, $(Xu) \cdot (Y - XB) = 0$ for all $u \in \mathbb{R}^p$. In other words, we have for all $u \in \mathbb{R}^p$,

$$\begin{aligned} 0 &= (Xu)^t (Y - XB) = u^t X^t (Y - XB) \\ &= u^t (X^t Y - X^t X B). \end{aligned}$$

Because $u^t (X^t Y - X^t X B) = 0$ for all $u \in \mathbb{R}^p$, it follows that $X^t Y - X^t X B = 0$. $\square$

2.6 In class exercises pt. II

1. Determine $X^t X$ and $X^t Y$ with

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1,p-1} \\ 1 & x_{21} & x_{22} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{n,p-1} \end{pmatrix}, \quad Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}.$$

2. Using (1) and by taking partial derivatives of

$$S(b_0, \dots, b_{p-1}) = \sum_{i=1}^n (y_i - (b_0 + b_1 x_{i1} + b_2 x_{i2} + \dots + b_{p-1} x_{i,p-1}))^2, \quad (2.5)$$

show that the hyperplane of best fit is obtained by solving $X^t X B = X^t Y$. (You could try this for $p = 3$ first.)

2.7 Nonlinear fittings

Although all of our examples so far have been linear fittings, we will demonstrate that least squares fittings works in the nonlinear case. What is important is that we have a candidate equation to fit. In the linear cases, we tried to fit

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_{p-1} x_{p-1}.$$

Suppose we have the following data as given in Figure 2.3. Instead of trying to fit the line $y = b_0 + b_1 x$, we could try to fit the parabola:

$$y = b_0 + b_1 x + b_2 x^2.$$

We can treat this the same way as before. Of course the quantities x and x^2 are *not* independent, but we can ignore this. Set

$$x_{i1} = x_i, \quad x_{i2} = x_i^2.$$

Therefore, the hyperplane of best fit for the data (x_{i1}, x_{i2}, y_i) will give us the parabola of best fit. *Try this on your own!*

So one can fit any hypersurface $y = f(x_1, \dots, x_{p-1})$ to the given data. The function f in this case is called the **regression function**. This general method of analysis is known as **regression analysis**. A few questions arise:

- Which surface is “best”?
- How can we quantify “best”?
- Even in the line case ($p = 2$), how can we quantify how well data fits our line?

2.8 Coefficient of determination (R^2 values)

We are going to make precise how well our hyperplane fits our data. Recall that hyperplanes can be replaced by hypersurfaces; see Section 2.7. First we establish some notation. Suppose we have n data points $(x_{i1}, x_{i2}, \dots, x_{i,p-1}, y_i) \in \mathbb{R}^p$. Then we define

$$\begin{aligned} \text{(Fitted value)} \quad & \hat{y}_i = b_0 + b_1 x_{i1} + b_2 x_{i2} + \dots + b_{p-1} x_{i,p-1}, \\ \text{(Residual)} \quad & e_i = y_i - \hat{y}_i, \end{aligned}$$

x_i	y_i
2.27	2.50
5.06	-16.13
1.45	4.23
5.89	-22.46
0.48	1.37
-0.22	0.86
1.44	11.85
-1.77	-14.71
2.45	9.42
-1.54	-14.07
7.55	-55.62
1.76	4.45
5.16	-19.56
3.26	-2.79
3.23	5.20
0.85	8.09

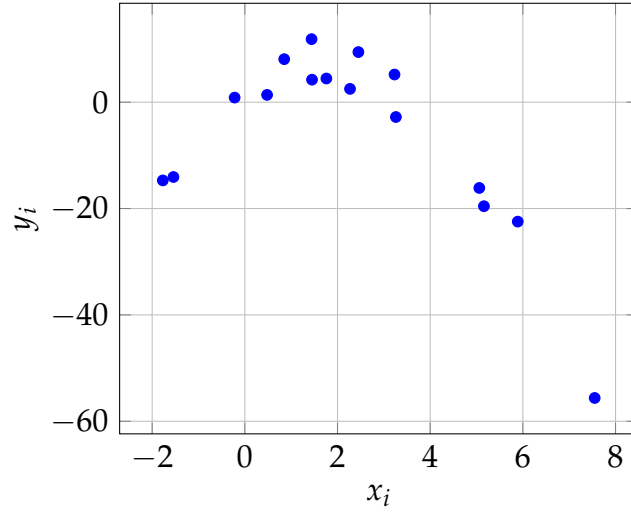


Figure 2.3: Data points demonstrating a nonlinear relationship.

(Sample mean) $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$

These yield vectors in $\mathbb{R}^n$ as follows

$$\hat{Y} = \begin{pmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{pmatrix} = XB, \quad E = \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = Y - \hat{Y}, \quad \bar{Y} = \begin{pmatrix} \bar{y} \\ \vdots \\ \bar{y} \end{pmatrix} = \bar{y} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

From our n data points, we have three points in $\mathbb{R}^n$ given by Y , $\hat{Y}$, and $\bar{Y}$. Three points always lie on a plane, so the three points determine a triangle on such a plane. What does this triangle look like? If it is a triangle (and not a line or a single point), then the next lemma proves it must be a right triangle.

Lemma 2.5. *If $X^tXB = X^tY$, then $E = Y - \hat{Y}$ and $\hat{Y} - \bar{Y}$ are orthogonal.*

Proof. Suppose $X^tXB = X^tY$. We need to prove two equations. For the first,

$$0 = X^t(Y - XB) = X^t(Y - \hat{Y}) = X^tE.$$

Hence, $X^tE = 0$. For the second,

$$\begin{aligned} \bar{Y}^tE &= \bar{y} \sum_{i=1}^n (y_i - \hat{y}_i) \\ &= \bar{y} \sum_{i=1}^n (y_i - (b_0 + b_1x_{i1} + b_2x_{i2} + \cdots + b_{p-1}x_{i,p-1})) \end{aligned}$$

$$= -\frac{\bar{y}}{2} \cdot \frac{\partial S}{\partial b_0} = 0,$$

where S is defined in (2.5), so $\bar{Y}^t E = 0$. Thus, we have

$$(\hat{Y} - \bar{Y}) \cdot E = (XB)^t E - \bar{Y}^t E = 0. \quad \square$$

Remark 2.6. One can simplify the proof for Lemma 2.5 by applying an isometry to the data, so that $\bar{y} = 0$. That is, one only needs to prove that E and $\hat{Y}$ are orthogonal.

The lengths of the differences of the vectors are important and have names:

$$\begin{aligned} (\text{Sums of Squares Total} - SST) &: \|Y - \bar{Y}\|^2, \\ (\text{Sums of Squares Error} - SSE) &: \|Y - \hat{Y}\|^2, \\ (\text{Sums of Squares Regression} - SSR) &: \|\hat{Y} - \bar{Y}\|^2. \end{aligned}$$

The value SST measures the *total variability* of the data set. For example, $\sqrt{SST} = \|Y - \bar{Y}\|$ is the distance from the actual data Y to the sample mean $\bar{Y}$. Using the same ideas, we can see that SSE measures the error of our regression and that SSR measures the distance from our regression to the sample mean.

Proposition 2.7.

$$SST = SSE + SSR.$$

Proof. Apply Lemma 2.5 and the Pythagorean Theorem:

$$\|Y - \bar{Y}\|^2 = \|Y - \hat{Y}\|^2 + \|\hat{Y} - \bar{Y}\|^2. \quad \square$$

Now we can describe a quantity that measures how good our regression fits the given data.

Definition 2.8. The **coefficient of determination** (also known as the R^2 -value) is

$$R^2 = \frac{SSR}{SST} = \frac{\|\hat{Y} - \bar{Y}\|^2}{\|Y - \bar{Y}\|^2}.$$

Proposition 2.9. $0 \leq R^2 \leq 1$.

Proof. Since each SST and SSR are squares, they are non-negative. By Proposition 2.7, we have $0 \leq SSR \leq SST$. $\square$

2.8.1 What do the extremes means?

The one case where R^2 is meaningless is when $SST = 0$. This implies both $SSR = SSE = 0$. Moreover, $Y = \bar{Y} = \bar{y}\mathbb{1}$, where $\mathbb{1}$ is the all ones column vector. Hence, every data point y_i is the same and, therefore, equal to the mean. Let's never return to this case.

We can have $SSR = 0$, which is equivalent to $R^2 = 0$. This implies that $\|\hat{Y} - \bar{Y}\|^2 = 0$, so that $\hat{Y} = \bar{Y}$. In other words, our prediction $\hat{y}_i$ is just simply the mean. This means we have not found any relationship between the independent variables and the dependent variables.

In the other extreme we have $SSR = SST$, which is equivalent to $R^2 = 1$. This implies that $Y = \hat{Y}$, so the given data lies (exactly) on the surface given by $y = f(x_1, \dots, x_{p-1})$. That is, the regression function exactly predicts the data.

To summarize, when $R^2 = 0$, we cannot deduce any relationship between the independent and dependent variables, and when $R^2 = 1$, we understand completely the relationship between the independent and dependent variables. Very roughly speaking, the R^2 can be thought of as the ratio of how well the regression fits the data.

2.9 In class exercises pt. III

1. Prove the following.

- (a) $\|\bar{Y}\|^2 = n\bar{y}$.
- (b) $Y \cdot \bar{Y} = \hat{Y} \cdot \bar{Y} = \|\bar{Y}\|^2$.
- (c) $Y \cdot \hat{Y} = \|\hat{Y}\|^2$.

2. Use (1) to show that

- (a) $SST = \|Y\|^2 - \|\bar{Y}\|^2$,
- (b) $SSE = \|Y\|^2 - \|\hat{Y}\|^2$,
- (c) $SSR = \|\hat{Y}\|^2 - \|\bar{Y}\|^2$.

3. What are the R^2 values for the examples above?

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